

- Capacity planning
 - Efficient capacity planning based on workloads and dynamic system changes
- Optimising resource allocation
 - Overrunning tasks
 - Long build times
- Optimising workflows
 - Creation of workflows aligned to the needs of the business

Challenges of implementing AI/ML in DevOps

Leveraging AI/ML in DevOps should be a planned and gradual approach to avoid common pitfalls. Some of the most common challenges include:

- An AI/ML model trained with improper data can lead to inaccurate

results and predictions undermining overall system efficiency.

- DevOps pipelines are hugely customised and there are a plethora of tools available in the market. The AI/ML model that is suitable for some tools may not work well when the organisation migrates to a different set of tools and technologies.
- AI/ML models may need continuous updates to keep with the agility of the processes, leading to frequent updates and added efforts.

The integration of AI/ML in DevOps is one of the key elements of digital transformation. This can rapidly enhance the capabilities of the DevOps processes and bring perceivable value to business. However, AI/ML adoption in DevOps is still evolving and possibilities of failure because of lack of proper implementation still exist. A good business vision, commitment from leadership, and the right change management strategy are essential for the successful implementation of AI/ML in DevOps. **END** 

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- **When traditional IT is better**
Traditional IT is more suitable for a healthcare organisation as stability and reliability are paramount to ensure patient safety and regulatory compliance. A formal change management process is required to mitigate risks in a sensitive healthcare environment. A risk-averse approach is preferred to prioritise patient well-being.

Considerations for choosing an approach

Several factors influence the decision.

- **Organisational size and complexity:** Larger organisations may benefit from DevOps to streamline operations, while smaller ones may find traditional IT sufficient.
- **Resource availability:** Implementing DevOps may require significant investments in tools and training.
- **Industry and compliance requirements:** Traditional IT aligns with industries with strict regulatory demands, such as healthcare.
- **Organisational culture:** The willingness to embrace change and

collaboration in adopting DevOps is a consideration.

- **Hybrid approach:** In some cases, a hybrid approach that combines elements of both DevOps and traditional IT can offer a balanced solution. Organisations can leverage the strengths of each approach to suit their specific needs.

A stockbroker organisation may adopt a hybrid approach by applying DevOps principles for rapid trading platform updates and customer-centric features, while maintaining traditional IT practices for stability in the core infrastructure.

A healthcare organisation could adopt a hybrid approach by using DevOps for non-critical applications, enabling rapid deployment of patient engagement tools, while maintaining traditional IT for critical healthcare systems, ensuring stability and compliance.

The comparison between DevOps and traditional IT in the context

of a stockbroker and a healthcare organisation highlights the importance of tailoring IT management approaches to specific industry needs and objectives. DevOps offers speed, collaboration, and customer focus, making it an attractive option for organisations seeking innovation and adaptability. Traditional IT prioritises stability and reliability, making it more suitable for industries with stringent compliance requirements and risk-averse environments.

Ultimately, successful IT management for any organisation requires thoughtful consideration of its unique context and a careful balance between the strengths of DevOps and a traditional IT-like hybrid approach. By understanding the benefits and limitations of each approach, organisations can make informed decisions that optimise software development, IT operations, and ultimately, their ability to deliver value to their customers and stakeholders. **END** 

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